TÍN HIỆU VÀ HỆ THỐNG LIÊN TỤC TRONG MIỀN THỜI GIAN

CHƯƠNG 2 – P1B

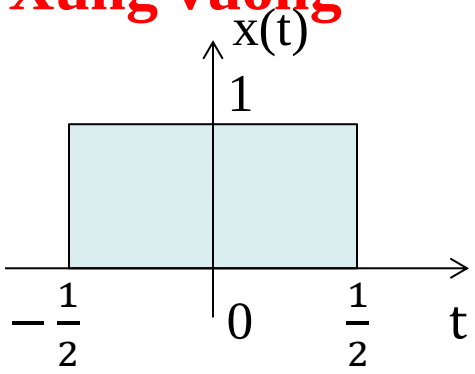
FEEE
CCE

Nội dung:

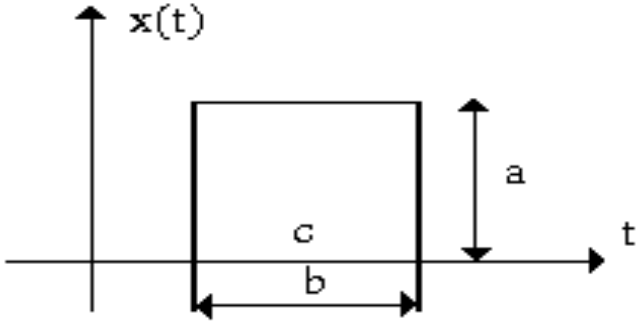
- Các thông số đặc trưng của tín hiệu.
- Một số tín hiệu xác định, thực.
- Tín hiệu phân bố.
- Phân tích tương quan.
- Tích chập.
- Hệ thống liên tục trong miền thời gian.

2.2 Một số tín hiệu xác định thực

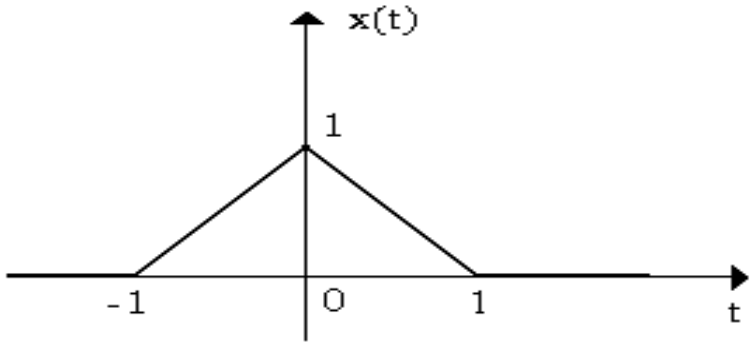
2.2.1 Xung vuông



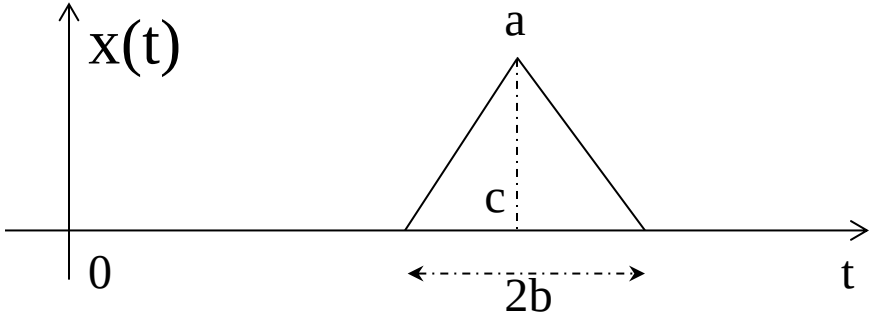
$$x(t) = \prod(t) = \begin{cases} 0 & \text{khi} \quad |t| > {}^{1}/_{2} \\ {}^{1}/_{2} & \text{khi} \quad |t| = {}^{1}/_{2} \\ 1 & \text{khi} \quad |t| < {}^{1}/_{2} \end{cases}$$



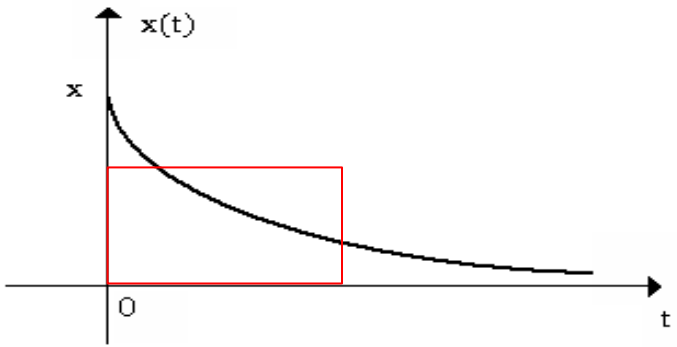
2.2.2 Xung tam giác



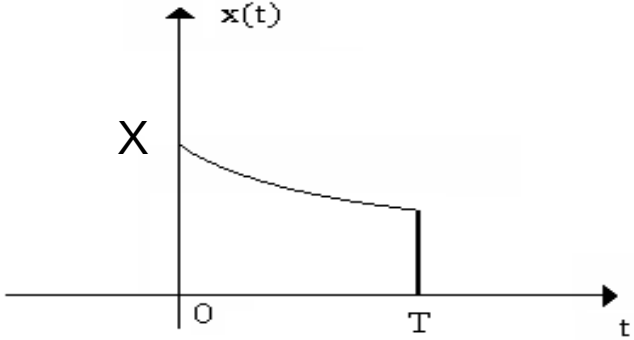
$$x(t) = \Lambda(t) = \begin{cases} 0 & khi\,|t| > 1 \\ 1-|t| & khi\,|t| \le 1 \end{cases}$$



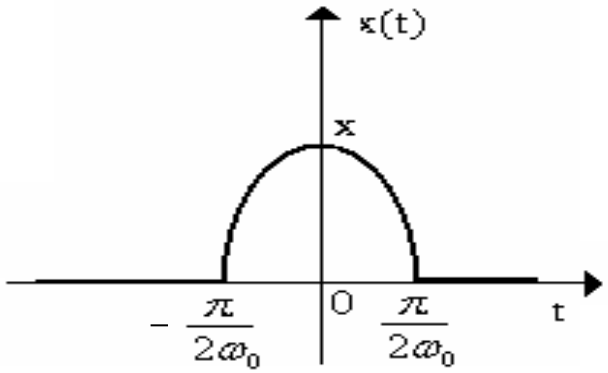
2.2.3 Xung hàm mũ



$$x(t) = X.e^{-\alpha t} \qquad \text{khi } \alpha > 0, \text{t} \geq 0$$

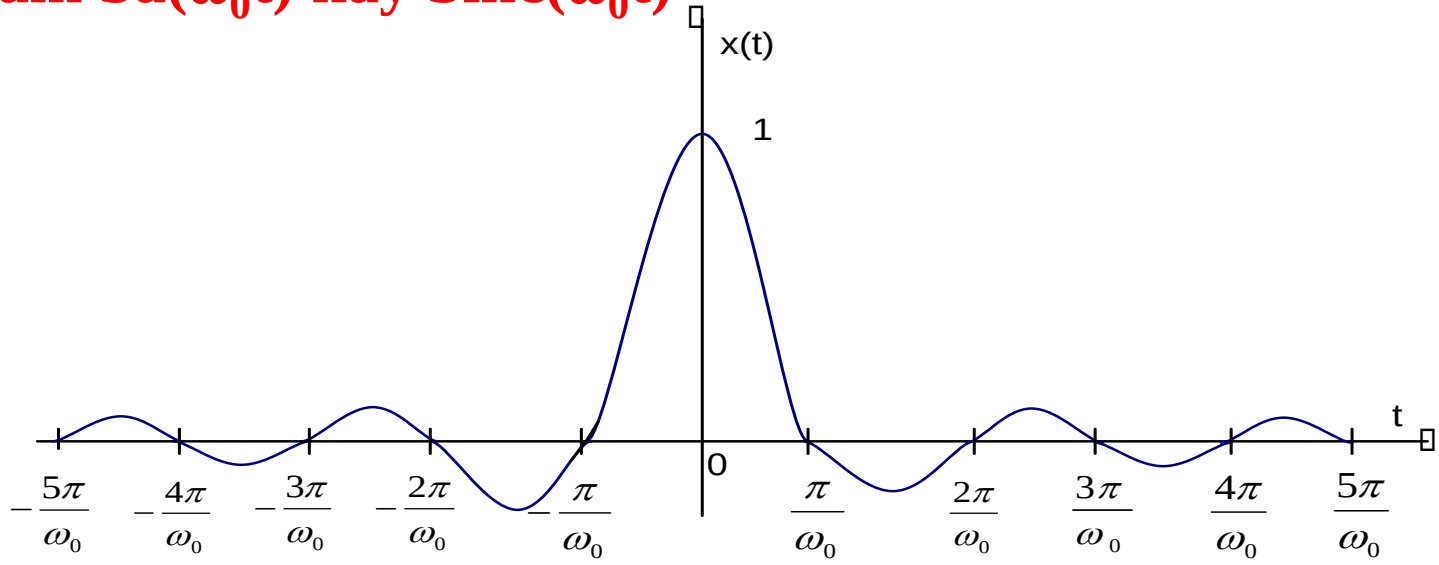


2.2.4 Xung cosin



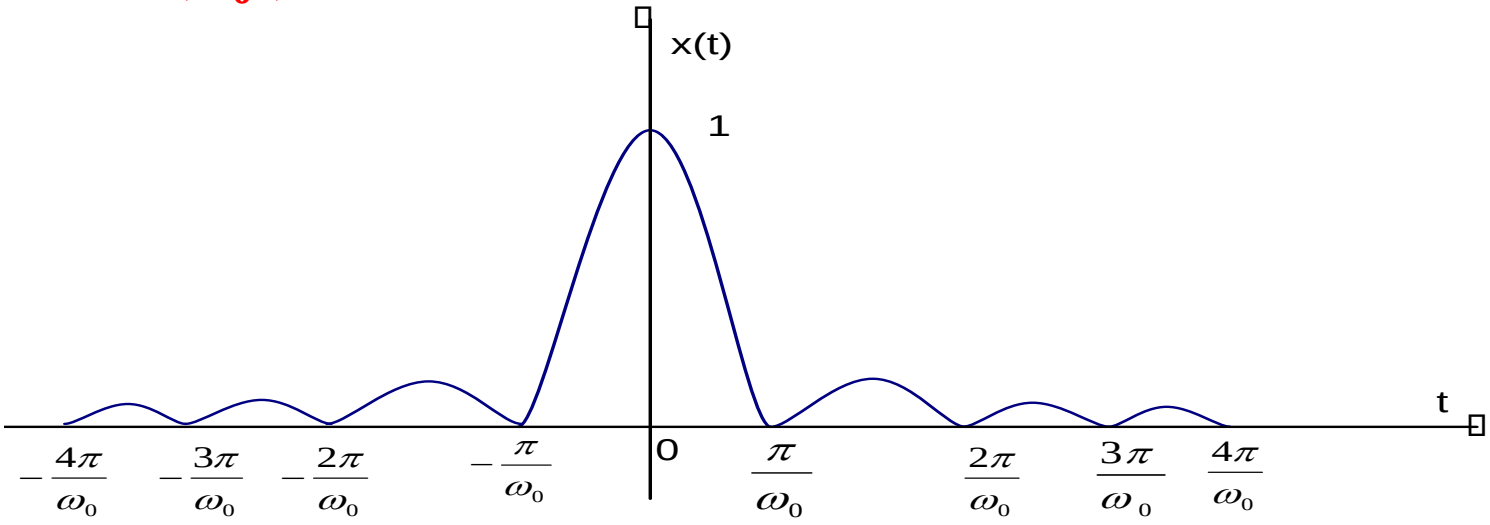
$$x(t) = \begin{cases} X.Cos(\omega_0 t) & \text{khi } |t| \le \dfrac{\pi}{2\omega_0} \\ 0 & \text{khi } t \text{ khác} \end{cases}$$

2.2.5 Hàm Sa(ω_0t) hay Sinc(ω_0t)



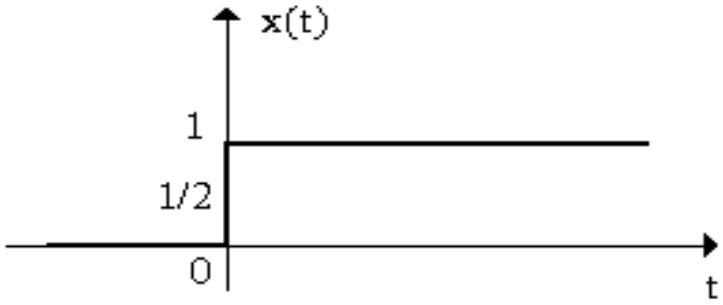
$$x(t) = Sa(\omega_0 t) = \begin{cases} \dfrac{\sin(\omega_0 t)}{\omega_0 t} & \text{khi } t \neq 0 \\ 1 & \text{khi } t = 0 \end{cases}$$

2.2.6 Hàm $Sa^2(\omega_0 t)$

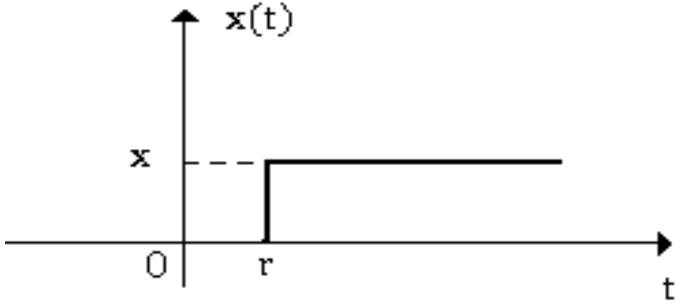


$$x(t) = Sa^2(\omega_0 t) = \begin{cases} \dfrac{\sin^2(\omega_0 t)}{(\omega_0 t)^2} & \text{khi } t \neq 0 \\ 1 & \text{khi } t = 0 \end{cases}$$

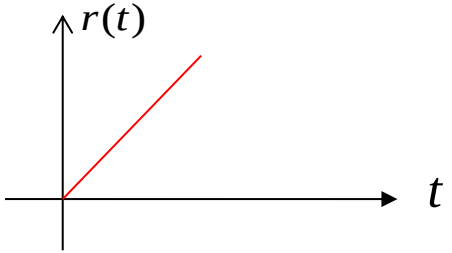
2.2.7 Hàm bước



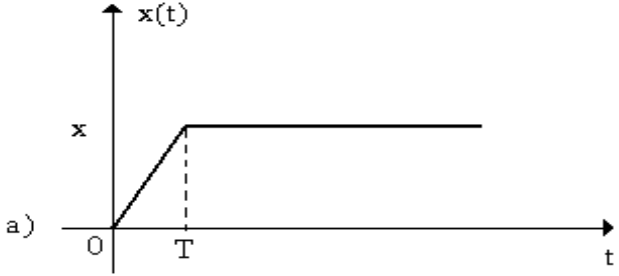
$$x(t) = u(t) = 1(t) = \begin{cases} 1 & \text{khi} \quad t > 0 \\ 1/2 & \text{khi} \quad t = 0 \\ 0 & \text{khi} \quad t < 0 \end{cases}$$

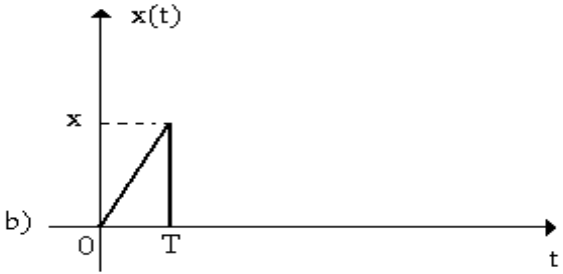


2.2.8 Hàm dốc

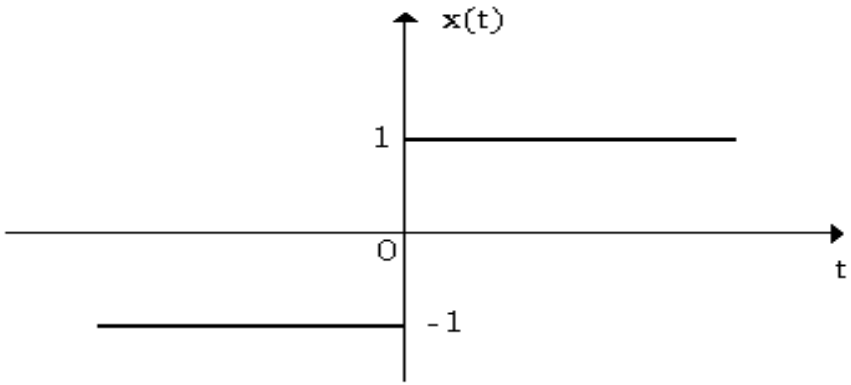


$$x(t) = r(t) = \begin{cases} t & t \geq 0 \\ 0 & t < 0 \end{cases}$$



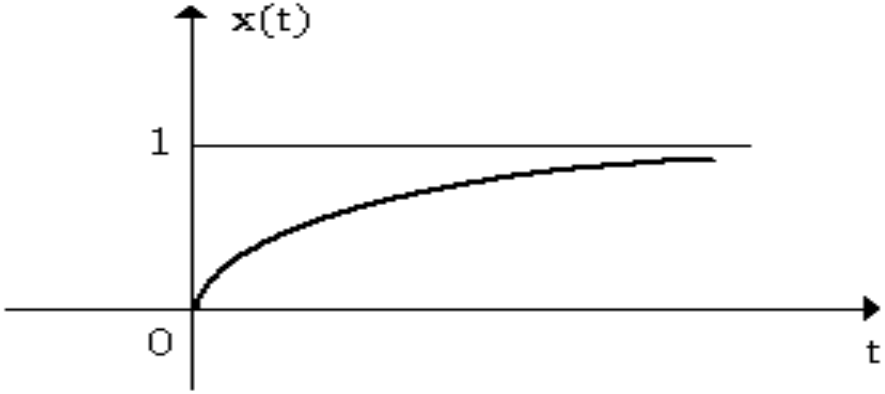


2.2.9 Hàm dấu



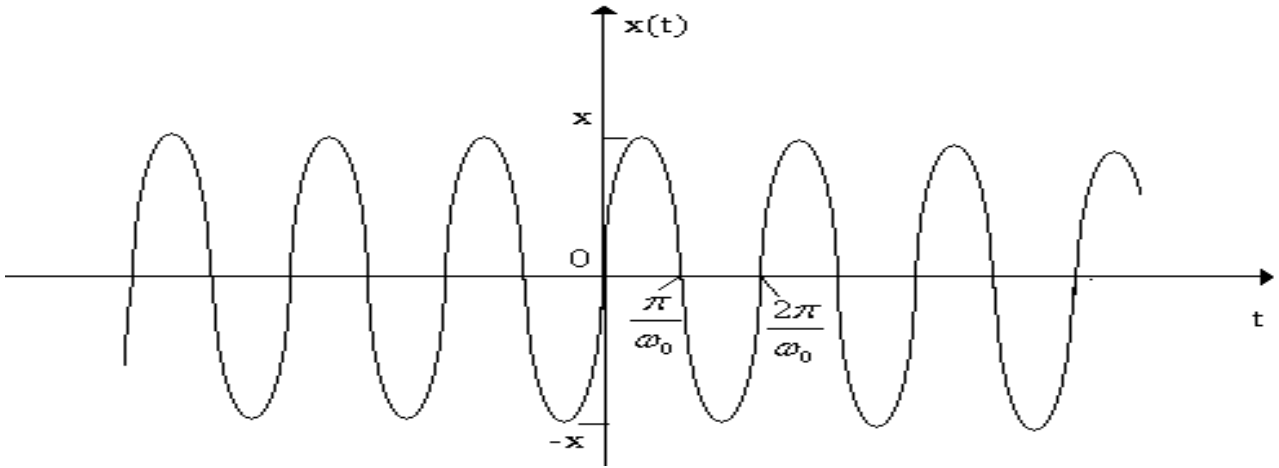
$$x(t) = \text{sgn}(t) = \begin{cases} 1 & \text{khi } t > 0 \\ 0 & \text{khi } t = 0 \\ -1 & \text{khi } t < 0 \end{cases}$$

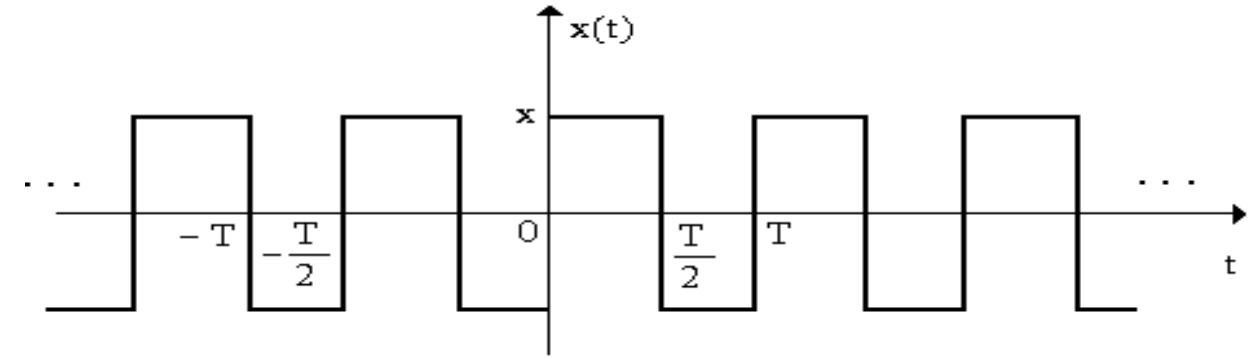
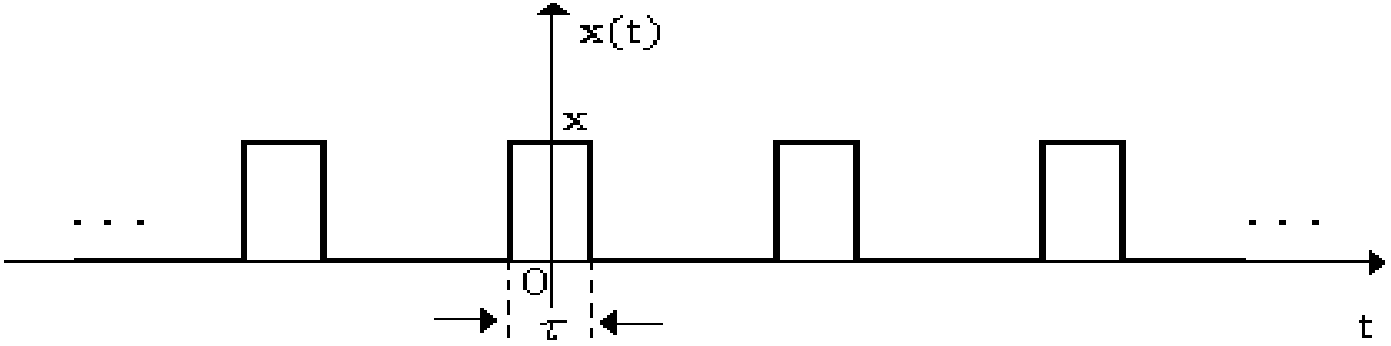
2.2.10 Xung hàm mũ



$$x(t) = 1 - e^{-\alpha t} \qquad \text{khi } \alpha > 0, \text{t} \geq 0$$

2.2.11 Tín hiệu điều hòa



$$x(t) = X.Sin(\omega_0 t)$$

2.2.12 Chuỗi xung vuông